Md. Jubayer Hossain

Bioinformatics & Computational Biology · Prospective Ph.D. Student

hossainmj.me@gmail.com · mdjubayerhossain.com · linkedin.com/in/hossainmj · github.com/hossainlab

EDUCATION

Master of Science in Microbiology 2019 – 2020
Jagannath University Dhaka, Bangladesh

Bachelor of Science in Microbiology 2016 – 2019
Jagannath University Dhaka, Bangladesh

CERTIFICATIONS & HACKATHONS

scverse × Cell Painting Hackathon 2026
scverse Consortium Berlin, Germany

Interdisciplinary Computational Biology 2026
BRAC University Dhaka, Bangladesh

AI for Public Health 2023
Child Health Research Foundation (CHRF) Dhaka, Bangladesh

AI in Public Health 2022
DeepLearning.AI Online Specialization

RESEARCH EXPERIENCE

Founder & Executive Director Jun 2020 – Present
CHIRAL Bangladesh (Institute for Volunteering Research) Dhaka, Bangladesh

- Founded an independent voluntary research institute and established national and international research partnerships.
- Instructed 3,000+ students in research methodology and bioinformatics through virtual training programs.
- Managed 50+ volunteer undergraduate research assistants on active computational biology and public health projects.
- Led collaborative teams to produce 20+ peer-reviewed publications with undergraduate co-authors.

Visiting Scholar Sep 2025 – Present
Dept. of Public Health, Daffodil International University Dhaka, Bangladesh

- Developed strategic collaborative initiatives to expand public health research output.
- Planned and executed epidemiological workflows to investigate emerging community health challenges.
- Wrote 5 peer-reviewed collaborative articles published in Q1/Q2 international journals.

Research Collaborator Sep 2026 – Present
Kyung Hee University Seoul, South Korea
Collaborator: Yeon Ju Kim, Ph.D.

Project 1: Single-Cell Transcriptomic Identification of Alzheimer's Disease Associated Molecular Targets and CNS–Peripheral Immune Signatures for Experimental Validation and Therapeutic Intervention

Project 2: Single-Cell Multi-Organ Profiling to In Vivo Validation: Identification of Hyperglycemia-Associated Gene Signatures and Their Modulation by GV1 in ObT2D Mice

- Developed single-cell workflows to uncover cell-type-specific targets prioritized for wet-lab validation.
- Integrated bulk RNA-seq and machine learning models to characterize perturbed pathways and disease signatures.
- Taught single-cell bioinformatic workflows using Python and R to lab members of Professor Kim.

Research Collaborator 2026 – Present
Lamar University Beaumont, TX, USA
Collaborator: Iqbal Mahmud, Ph.D.

Project 1: EWSAtlas: A Harmonized Single-Cell Transcriptomic Atlas of Ewing Sarcoma

Project 2: Unified Single-Cell Atlas of Diffuse Midline Glioma (DMG)

- Developed single-cell pipelines to harmonize multi-cohort transcriptomic atlases for Ewing sarcoma (EWSAtlas) and diffuse midline glioma (DMG).
- Characterized neurodevelopmental lineage stalling programs, malignant cell plasticity, and microenvironmental niches.
- Translated computational findings into oncogenic mechanisms and prioritized therapeutic targets.

Research Collaborator 2024 – Present
One Health Lab, Jahangirnagar University Dhaka, Bangladesh
Collaborator: Md. Salequl Islam, Ph.D.

Project 1: Genomic Profiling of Extended-Spectrum Antimicrobial Resistance Genes (ARGs) and Mobilizable Resistance Determinants in Industrial Effluents

Project 2: One Health Assessment of Antimicrobial Usage, Microbial Dynamics, and Resistance Transmission Pathways Across Poultry Farm Environments and Farm Workers in Bangladesh

- Profiled whole-genome sequencing (WGS) and metagenomic assemblies to identify resistance genes in wastewater.
- Developed reproducible workflows for resistome profiling, quantifying horizontal gene transfer risks in effluents.
- Instructed lab members of Professor Islam in reproducible bioinformatics workflows and genomic data analysis.

Research Intern Feb 2022 – Jun 2022
International Center for Diarrhoeal Disease Research, Bangladesh Dhaka, Bangladesh
Advisor: Dr. Kamrun Nahar Koly

Project: Systematic Review and Evidence Synthesis in Intimate Partner Violence

- Formulated systematic review protocols following PRISMA guidelines to evaluate violence prevention interventions.
- Screened and cataloged over 8,000 publications across PubMed, Scopus, and Web of Science using structured criteria.
- Extracted quantitative and qualitative epidemiological indicators into standardized databases for synthesis.
- Assessed methodological quality and risk of bias across observational studies to ensure evidence synthesis rigor

Data Enumerator Oct 2022 – Nov 2022
School of Allied Health, Curtin University Australia
Collaborator: Bau DA Madjid

Project: Delphi Study on Patient Outcomes Measurement for Interprofessional Tuberculosis Care in Bangladesh

- Coordinated national-level execution of an international Delphi study evaluating interprofessional tuberculosis care in Bangladesh.
- Facilitated consensus gathering across iterative survey rounds, securing responses from 25 clinical specialists.

PROFESSIONAL AND WORK EXPERIENCE

Founder & CEO Jan 2025 – Present
DeepBio Limited Dhaka, Bangladesh

- Founded and led DeepBio, applying foundational AI models to regional biological and healthcare challenges.
- Designed Project PROTIRODH to predict and rescue doomed beta-lactams against hospital-acquired infections.
- Established collaborations with Bangladeshi universities for experimental wet-lab validation.
- Launched ”DeepBio Academy”, upskilling 1,000+ undergraduates in bioinformatics and AI in life sciences.
- Launched ”Agami AI”, leading national AI literacy programs for 1,000+ school and college students.

PUBLICATIONS

Underline indicates mentee; *corresponding

A. Journal Articles

1. Islam, M. W., Rahman, M. M., Naznin, H., Hossain, M. S., Akter, T., Shatabde, Z. A., & *__Hossain, M. J.__ (2026) “Integrative Bioinformatics Analysis Reveals COL13A1 and COL23A1 as Potential Diagnostic and Prognostic Biomarkers in Thyroid Cancer” *Health Science Reports*. DOI: 10.1002/hsr2.72617
2. *__Hossain, M. J.__, Shahariar, M., Barsha, LHJ. Shahjahan, M., Towhid, ST., Sheikh, MK., Hasan, MM., Rahman, MH.,

Sazid, MS. (2026) "Lack of knowledge and training about antibiotic resistance among community pharmacists in Bangladesh: a cross-sectional study" *Health Science Reports*. DOI: 10.1002/hsr2.72866

3. *__Hossain, M. J.__, Afrin Mim, N., Akter, N., Akter Sony, S., Haque Saikat, E., Al-Amin Hossain, M., Parvin, N., Evan, M. R., Jannat Juma, J., Parmita Roy, P., Islam Maruf, M. F., Islam, R., Labib, A., Akter Shatabde, Z., Das, M., & Amanulla, M. (2026) "Epidemiological Trends, Public Health Challenges, and Strategic Control Priorities of Dengue in Bangladesh (2000—2024): A Narrative Review" *Health Science Reports*. DOI: 10.1002/hsr2.72747
4. Fariha, F. T. J., Fuad, M., Saha, C. S., Hossen, S., & *__Hossain, M. J.__ (2025) "Comprehensive bioinformatics analysis reveals prognostic significance and immunological roles of WNT gene family in breast cancer" *Scientific Reports*. 15, 34490. DOI: 10.1038/s41598-025-13315-6
5. Ahmed, M. Z., Billah, M. M., Ferdous, J., & *__Hossain, M. J.__ (2025) "Pan-cancer analysis reveals immunological and prognostic significance of CCT5 in human tumors" *Scientific Reports*. 15, 14405. DOI: 10.1038/s41598-025-88339-z
6. Bari, S. M., Fuad, M., __Hossain, M. J.__, Ahammad, I., Begum, M., & Helal, M. M. U. (2025) "A meta-analysis of public RNA-Seq data identifies conserved stress responses in rainbow trout" *BMC Genomics*. 26, 999. DOI: 10.1186/s12864-025-12127-2
7. *__Hossain, M. J.__, Das, M., Shahjahan, M., Islam, M. W., & Towhid, S. T. (2025) "Clinical and hematological manifestation of dengue patients in 2022 outbreak" *Health Science Reports*. 8, e70356. DOI: 10.1002/hsr2.70356
8. Das, M., & __Hossain, M. J.__ (2025) "Young stroke in Bangladesh: Addressing rare cases with diagnostic challenges" *Stroke and Vascular Neurology*. svn-2025-004178. DOI: 10.1136/svn-2025-004178
9. *__Hossain, M. J.__, Sony, S. A., Fariha, F. T. J., & Hossen, S. (2025) "Preventing the silent threat: A perspective on preparing Bangladesh for Human Metapneumovirus" *Health Science Reports*. 8, e71101. DOI: 10.1002/hsr2.71101
10. Shanta, A. S., Islam, N., Al Asad, M., Akter, K., Habib, M. B., __Hossain, M. J.__, Nahar, S., Godman, B., & Islam, S. (2024) "Resistance and co-resistance of metallo-beta-lactamase genes in diarrheal and urinary-tract pathogens" *Microorganisms*. 12(8), 1589. DOI: 10.3390/microorganisms12081589
11. *__Hossain, M. J.__, Das, M., & Munni, U. R. (2024) "Urgent call for compulsory premarital screening for thalassemia prevention in Bangladesh" *Orphanet Journal of Rare Diseases*. 19, 326. DOI: 10.1186/s13023-024-03344-1
12. Islam, M. W., Shahjahan, M., Azad, A. K., & *__Hossain, M. J.__ (2024) "Factors contributing to antibiotic misuse among parents in Dhaka City" *Scientific Reports*. 14, 2318. DOI: 10.1038/s41598-024-52313-y
13. *__Hossain, M. J.__, Das, M., Islam, M. W., Shahjahan, M., & Ferdous, J. (2024) "Community engagement and social participation in dengue prevention" *Health Science Reports*. 7, e2022. DOI: 10.1002/hsr2.2022
14. *__Hossain, M. J.__, Azad, A. K., Shahid, M. S. B., Shahjahan, M., & Ferdous, J. (2024) "Prevalence, antibiotic resistance pattern for bacteriuria from patients with UTIs" *Health Science Reports*. 7, e2039. DOI: 10.1002/hsr2.2039
15. Akter, M. M., & *__Hossain, M. J.__ (2024) "Food consumption patterns and sedentary behaviors among university students" *Health Science Reports*. 7, e70259. DOI: 10.1002/hsr2.70259
16. *__Hossain, M. J.__, Islam, M. W., Munni, U. R., Gulshan, R., Mukta, S. A., Miah, M. S., Sultana, S., Karmakar, M., Ferdous, J., & Islam, M. A. (2023) "Health-related quality of life among thalassemia patients in Bangladesh (SF-36)" *Scientific Reports*. 13(1). DOI: 10.1038/s41598-023-34205-9
17. Towhid, S.T., __Hossain, M. J.__, Sammo, M.A.S., & Akter, S. (2022) "Perception of Students on Antibiotic Resistance and Prevention" *Eur. J. Bio. Biotech*. 3(3). DOI: 10.24018/ejbio.2022.3.3.341
18. __Hossain, M. J.__, Towhid, S.T., Sultana, S., Mukta, S.A., Gulshan, R., Miah, M.S. (2022) "Knowledge and Attitudes towards Thalassemia among Public University Students" *Microbial Bioactives*. 5(2). DOI: 10.25163/microbbioacts.526325

B. Papers Under Review

1. Kundu, P., Shahariar, M., Munny, F. A., Riya, A. T., Lota, H. B., Ahmed, M. Z., Meme, M. A., Kona, U. M. P., Mahmuda, F., Chowdhury, S. S., Sraboni, S. A., & *__Hossain, M. J.__ (2026) "Computational identification of Taxus baccata-derived phytochemicals targeting EZH2 to overcome therapeutic resistance in melanoma" *Scientific Reports*. DOI: 10.21203/rs.3.rs-8494315/v1
2. Islam, M. W., Fariha, F. T. J., Ahmed, M. Z., Ferdous, J., Lota, H. B., Hossain, M. S., Kundu, P., Shahjahan, M., & *__Hossain, M. J.__ (2025) "Bioinformatics-driven multi-omics profiling of CDK1 and CDK6 identifies prognostic and therapeutic roles in breast cancer" *Health Science Reports*.
3. Billah, M. M., Mabsurah, K., Ahammad, K., Yeana, I. J., Sumaiya, M., Islam, T., Bhattacharjee, A., Ferdous, J., & *__Hossain, M. J.__ (2025) "Multi-omics pan-cancer analysis reveals an immunological role and prognostic potential of WDR76" *BMC Genomics*.
4. Sony, S. A., Kundu, L. R., Limon, M. H., Chowdhury, T. B. K., & & *__Hossain, M. J.__ (2025) "Predicting early antenatal care initiation at the first trimester among reproductive women in Bangladesh using machine learning" *BMC Pregnancy and Childbirth*.

C. Preprints

1. **Hossain, M. J.** (2026) "metaXpress: An End-to-End Bioconductor Framework for Quality Control, Harmonization, and Multi-Cohort Bulk RNA-seq Meta-Analysis" *bioRxiv*. GitHub: github.com/hossainlab/metaXpress
2. **Hossain, M. J.** (2026) "Graph Neural Network for predicting immunosuppressive spatial niches in glioblastoma from 10x Visium spatial transcriptomics" *bioRxiv*. GitHub: github.com/hossainlab/deepgraph-gbm
3. **Hossain, M. J.** (2026) "Translating Perturb-seq Screens Across Cell Lines" GitHub: github.com/hossainlab/perturb-bench
4. **Hossain, M. J.**, & Dev, P.C. (2026) "Evaluating the false-positive rate of single-cell copy-number and malignant-cell classifiers" *bioRxiv*. GitHub: github.com/hossainlab/scCNV-FP-Evaluation
5. **Hossain, M. J.**, Abhi, N. I., Islam, M. M., Das, S., Tasneem, E., & Tasnim, M. (2026) "Dosage-dependent disruption of corticofugal neuron development by Fezf2 loss" *bioRxiv*. GitHub: github.com/hossainlab/fezf2-loss-multi-omics
6. **Hossain, M. J.**, Dev, P. C., Nawar, T., Arony, T. A., Hoque, M., Chowdhury, D., & Rahman, M. A. (2026) "A donor-level map of immune sex differences" *bioRxiv*. GitHub: github.com/hossainlab/pmc_immune_landscape

D. Working Papers

1. **Hossain, M. J.**, Dev, P.C., & Shahariar, M. (2026) "Deep Learning-Driven AMR Drug Repurposing" *Working Paper*. GitHub: github.com/hossainlab/amr_drug_repurposing

PRESENTATIONS

Underline indicates mentee; *corresponding

A. Invited Speaker

1. **Hossain, M. J.** Undergraduate Research - Importance, Benefits, and Challenges. CHIRAL Bangladesh, Dhaka, Bangladesh, March, 2019.
2. **Hossain, M. J.** State the Art of Microbial Genome Analysis. Jagannath University, Dhaka, Bangladesh, May, 2018.
3. **Hossain, M. J.** Computational Biology and Bioinformatics Research in Resource-Limited Settings: Strategies, Tools and Opportunities. Jagannath University Higher Study and Research Society, Dhaka, Bangladesh, June, 2015.
4. **Hossain, M. J.** Mastering Biomedical Data Management. IFMSA Bangladesh, Dhaka, Bangladesh, April, 2014.

B. Conference Presentations

1. Rahman, M. A., Shahjahan, M. and *__Hossain, M. J.__ EBV-antigen-specific T cell clonotype fate in nasopharyngeal carcinoma: A single-cell TCR analysis. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh, December, 2026.
2. Chowdhury, D., Shahjahan, M. and *__Hossain, M. J.__ Reassessing the chemotherapy–TIGIT association in pancreatic cancer through cross-cohort validation. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh, December, 2026.
3. Hoque, M., Shahjahan, M. and *__Hossain, M. J.__ Chemotherapy-induced tumor cell state dynamics in osteosarcoma: NMF-based decomposition and independent bulk validation. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh, December, 2026.
4. Arony, T. A., Shahjahan, M. and *__Hossain, M. J.__ Spatial mapping of diffuse midline glioma cell states reveals conserved territorial architecture and a mesenchymal-like interface for tumor–microenvironment crosstalk. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh, December, 2026.
5. Nawar, T., Shahjahan, M., *__Hossain, M. J.__ Cross-Cohort Meta-Analysis of Acral Melanoma Immune Cell Quantity vs. Function With Three-Way Decomposition. Dhaka University Research Society (DURS), University of Dhaka, Bangladesh, December, 2026.
6. Islam Abhi, S. N., Meme, M. A., Shahariar, M., Barsha, L., *__Hossain, M. J.__ Single-cell transcriptomic and network pharmacology insights into the therapeutic potential of Withania somnifera-derived phytochemicals against cisplatin-resistant ovarian cancer. BEICTCON2026, IEEE, IBUAT, Dhaka, Bangladesh, August, 2026.
7. Islam, M. M., *__Hossain, M. J.__ Fezf2-Mediated Cortical Development: Multi-Omics Single-Cell Analysis. Network of Young Biotechnologists Bangladesh, Dhaka, Bangladesh, April, 2026.

8. **Hossain, M. J.**, Manisha Das Diagnostic and Electrophysiological Features of Hirayama Disease in Young Adult Male: A Case Report. Institute of Epidemiology, Disease Control and Research (IEDCR), Dhaka, Bangladesh, May, 2020.
9. Nayeem, M.U., Mrittika, M.A., Azad, A.K., Ferdous, J., Ahmed, S., Sanyal, S.K., Towhid, S.T. and **Hossain, M. J.** Quantitative Microbial Risk Assessment from Vancomycin Resistant Enterococcus faecalis and Enterococcus faecium from a Specific Neighborhood in Dhaka City, Bangladesh. Bangladesh Society of Microbiologists, Dhaka, Bangladesh, May, 2019.

C. Posters

1. Ahmad, F., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Identification of Subtype-Specific Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis. Bangladesh Society of Microbiologists, University of Dhaka, Bangladesh, December, 2025.
2. Aditee, L. M., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma. Bangladesh Society of Microbiologists, University of Dhaka, Bangladesh, December, 2025.
3. Ahamed, M. S., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Bioinformatic Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer. Bangladesh Society of Microbiologists, University of Dhaka, Bangladesh, December, 2025.
4. Tabassum, R., Fuad, M., Shahjahan, M. and *__Hossain, M. J.__ Comparative Transcriptomic Meta-Analysis Reveals Shared and Distinct Transcriptional Networks and Pathway Crosstalk in Hepatocellular Carcinoma for Biomarker and Therapeutic Target Identification. Bangladesh Society of Microbiologists, University of Dhaka, Bangladesh, December, 2025.

TEACHING & MENTORING EXPERIENCE

A. Teaching & Training Roles

Faculty (Bioinformatics) Aug 2023 – Present
cBLAST, University of Dhaka Dhaka, Bangladesh

- Taught online workshops on biomedical machine learning with Python to 20+ participants.
- Developed self-paced courses on data science and machine learning for biologists on the cBLAST platform.
- Coordinated with Dr. Zeba Islam Seraj to advance accessible bioinformatics training initiatives.

Program Lead Jul 2025 – Present
GSA Bioinformatics Internship Dhaka, Bangladesh

- Designed the 6-month internship program and curriculum funded by ASI School of Life to train computational biologists.
- Mentored 10 interns across two cohorts in cancer genomics, bulk RNA-seq, and single-cell transcriptomics.
- Managed a multi-institutional team to mentor interns through independent research projects.

Curriculum Developer and Instructor Jan 2024 – Present
School of Bioinformatics Dhaka, Bangladesh

- Designed foundational bioinformatics curricula tailored for high school (grades 9–10) and college (grades 11–12) students.
- Taught introductory computational biology sessions across two student cohorts.
- Promoted early-stage bioinformatics literacy and scientific awareness among secondary and higher secondary students.

Teaching Assistant Oct 2022 – Dec 2022
Department of Microbiology, Jagannath University Dhaka, Bangladesh

- Delivered research methodology training to 40+ undergraduates and reviewed literature assessments.
- Reviewed and evaluated undergraduate research proposals, advising on study design.
- Guided undergraduate students data analysis practicals for the Public Health and Hygiene course

B. Courses Taught

Reproducible Bioinformatics Workflows: Ideation to Execution, DeepBio Academy [Materials] May 2026

Statistical Analysis with R for Research, Research Help Bangladesh [Materials] May 2026
Publication-ready Tables with R, CHIRAL Bangladesh [Materials] Apr 2025
Building Dashboard with R, CHIRAL Bangladesh [Materials] Apr 2025
Easystats for Biomedical Researchers, CHIRAL Bangladesh [Materials] Apr 2025
R for Bioinformatics, CHIRAL Bangladesh [Materials] Apr 2025
Data Analysis with R, CHIRAL Bangladesh [Materials] Apr 2025
R for Cancer Bioinformatics, CHIRAL Bangladesh [Materials] Apr 2025
R for Research, CHIRAL Bangladesh [Materials] Oct 2023
Machine Learning for Bioinformatics, CHIRAL Bangladesh [Materials] Aug 2023
RNA-Seq Analysis with R, CHIRAL Bangladesh [Materials] Aug 2023
AI for Drug Discovery, CHIRAL Bangladesh [Materials] Aug 2023
Cancer Bioinformatics, CHIRAL Bangladesh [Materials] Aug 2023
Python for Health Data Analytics, CHIRAL Bangladesh [Materials] Aug 2023
Statistical Analysis with R for Research, DeepBio Academy, CHIRAL Bangladesh [Materials] Oct 2014
Statistical Analysis with R for Research, Genetic Engineering Club, JnU [Materials] Oct 2014

MENTORING

1. **Dhruba Chowdhury** (2026), University of Dhaka: A Harmonized Single-Cell Atlas of Early-Stage, Treatment-Naïve Pancreatic Ductal Adenocarcinoma
2. **Md. Aftabur Rahman** (2026), Jahangirnagar University: A Multi-Cohort Single-Cell Atlas of H3-K27M Diffuse Midline Glioma Across Age, Anatomical Location, and Histone Variant
3. **Tahia Aswad Arony** (2026), North South University: Integrative Single-Cell Profiling of Human Frontotemporal Dementia Organoid Models Reveals Convergent Cellular Programs Across Genetic Subtypes
4. **Mahmuda Hoque** (2026), BRAC University: Building a Harmonized Single-Cell Reference Atlas of Osteosarcoma Reveals Conserved Tumour, Stromal, and Immune Cell States
5. **Shreya Tabassum** (2026), BRAC University: Tracing Organ-Specific Metastatic Programs in Lung Adenocarcinoma Through Harmonized Single-Cell Transcriptomic Integration
6. **Md Shakil Ahamed** (2025), Mawlana Bhashani Science and Technology University: Bioinformatics Analysis of Transcriptomic Data Reveals Molecular Signatures Driving Colorectal Cancer
7. **Rahnuma Tabassum** (2025), Jagannath University: Integrated Transcriptomic Meta-Analysis Identifies Pathway Crosstalk and Therapeutic Targets in Hepatocellular Carcinoma
8. **Fayez Ahmad** (2025), Notre Dame College, Dhaka: Identification of Prognostic Biomarkers in Esophageal Carcinoma through Integrated Bioinformatics Analysis
9. **Sabbir Khan** (2025), Gazipur Agricultural University: Multi-Omics Meta-Analysis Identifies a Mitotic Signature as a Prognostic and Therapeutic Target in KIRC
10. **Lamisa Manha Aditee** (2025), BRAC University: Network-Driven Bioinformatics Approach Uncovers Prognostic Signatures and Therapeutic Targets in Pancreatic Ductal Adenocarcinoma

AWARDS & FELLOWSHIPS

Open Bioinformatics Foundation Event Fellowship, *Open Bioinformatics Foundation* 2026
Travel Grant for scverse x cellpainting Hackathon, *scverse* 2026
Scholarship for AI in Public Health Workshop, *Bill and Melinda Gates Foundation* 2023
Dutch-Bangla Bank (DBBL) Merit Scholarship, *Dutch-Bangla Bank* 2013

COMMUNITY SERVICES

Life Member Sep 2026 – Present

Global Network of Bangladeshi Biotechnologists (GNOBB) Dhaka, Bangladesh

- Promoted national and international biotechnology networking to foster scientific exchange and partnerships.
- Facilitated collaborative research initiatives connecting Bangladeshi biotechnologists worldwide.
- Provided computational biology mentorship and professional guidance to emerging life science researchers.

Member Jan 2017 – Dec 2019
Bio-Bio-1:Bioinformatics Research Discussion Group Dhaka, Bangladesh

- Delivered technical bioinformatics workshops for student and researcher discussion groups.
- Contributed to mathematical modeling of chikungunya transmission dynamics and epidemiological analysis.
- Engaged in regular scientific discussions and peer review of computational biology research topics.

Nextflow Ambassador Jan 2025 – Present
Seqera Barcelona, Spain

- Advocated for scalable and reproducible bioinformatics workflows utilizing Nextflow and nf-core standards.
- Organized and delivered hands-on workflow engineering workshops for the Bangladeshi bioinformatics community.
- Mentored local researchers and students in adopting cloud computing and modern pipeline automation.

TECHNICAL SKILLS

- **Programming Languages**: Python, R, Julia, SQL, Bash, JavaScript
- **Bioinformatics**: Bulk RNA-Seq, Single-Cell RNA-Seq, Spatial Transcriptomics, Multi-omics
- **Workflow Management & Reproducibility**: Nextflow (DSL2), nf-core, Docker, Singularity, Git, GitHub
- **AI in Biology**: Scikit-learn, PyTorch, RDkit, DeepChem
- **HPC & Cluster Computing**: SLURM (sbatch, sarray), Nextflow DSL2 with SLURM executor for scalable reusable pipelines
- **Cloud**: AWS S3, Google Cloud Storage for large genomic datasets

RESEARCH INTERESTS

- **Bioinformatics**: Integrative analysis of bulk RNA-seq, single-cell RNA-seq, and spatial transcriptomics to identify robust immunological biomarkers and therapeutic targets within tumor microenvironments;
- **AI in Biology**: Deep learning applications in single-cell genomics, particularly for data integration and cell type annotation.

TEACHING INTERESTS

- **Programming & Data Analytics**: Programming in R and Python; exploratory data analysis; data visualization; reproducibility.
- **Bioinformatics**: Scalable, reproducible computational pipelines for single-cell, spatial, and bulk transcriptomics to elucidate complex disease biology (Bash, Nextflow, Docker).
- **AI in Biology**: Applying deep learning frameworks to single-cell genomics data for high-accuracy predictive modeling.

MEDIA COVERAGE

1. May 27, 2024, **The Business Standard**: Quoted in article What the rise in self-medication tells us about the country's healthcare system.
2. Feb 08, 2024, **The Business Standard**: Study featured in article "Parental lack of antibiotic knowledge imperils child health in Bangladesh: Study".

REFERENCES

Dr. Syeda Tasneem Towhid
Associate Professor
Department of Microbiology
Jagannath University
Dhaka, Bangladesh
towhidst@mib.jnu.ac.bd

Dr. Iqbal Mahmud
Assistant Professor
Lamar University
Beaumont, TX, USA
imahmud1@lamar.edu

Dr. Zeba Islam Seraj
Director, cBLAST
University of Dhaka
Dhaka, Bangladesh
zebai@du.ac.bd

Dr. Yeon Ju Kim
Professor
Convergent Biotechnology
Kyung Hee University
Seoul, South Korea
yeonjukim@khu.ac.kr